

PBMC Cell-Type Annotation

LLM Pipeline Comparison: Claude vs Llama vs Gemma

Zero-Shot (Marker Genes + LLM)

Reference Mapping (Pre-trained Atlas)

Dataset: PBMC 3k | AWS Bedrock | scanpy · CellTypist · Scanorama

Scripps Research Hackathon 2026

Claude Haiku 4.5

Llama 3.1 70B

Gemma 3 27B

Pipeline Design Comparison

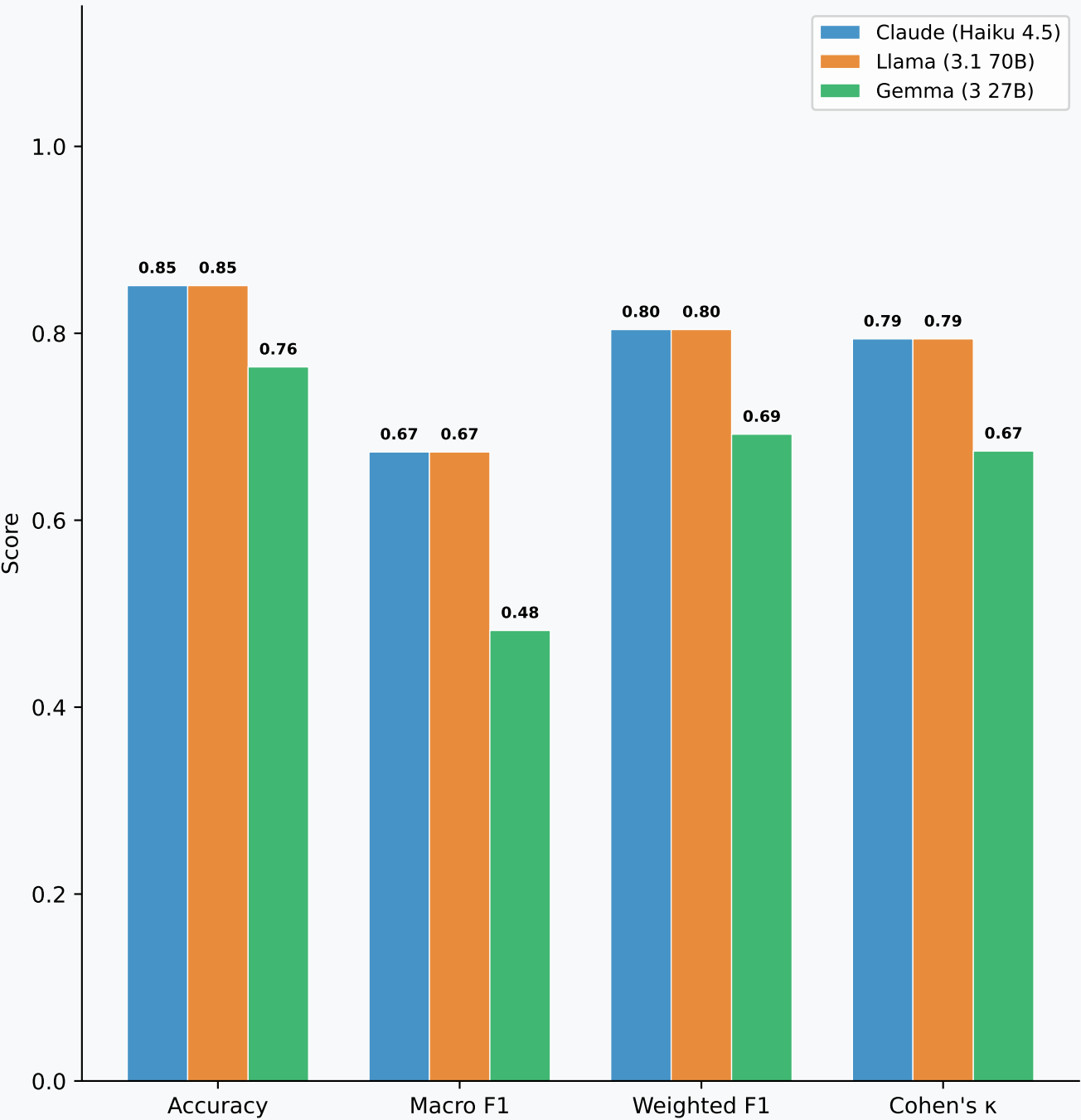
What did each LLM choose for each method?

	Claude (Haiku 4.5)	Llama (3.1 70B)	Gemma (3 27B)
Zero-Shot Approach	LLM API call via Bedrock (biological reasoning)	Average marker gene expression per cluster	Sum marker gene expression per cell
Zero-Shot Granularity	Cluster-level	Cluster-level	Per-cell
Reference Method	CellTypist (logistic regression)	Centroid dot-product similarity	Scanorama integration + KNN label transfer
Reference Dataset	Immune_All_High (700k cells, 20+ studies)	pbmc68k_reduced (700 cells, 10x Genomics)	pbmc68k_reduced (700 cells, 10x Genomics)
Code Structure	4 scripts (pipeline)	1 monolithic script	6 modular scripts
AWS Compute Suggested	EC2 (r5.xlarge)	EC2 + Batch	EC2 (m5.2xlarge)
Requires Bedrock API	✓ Yes (LLM call)	✗ No	✗ No

Zero-Shot Annotation Results

Cluster marker genes → cell type label (no reference data)

Metric Comparison



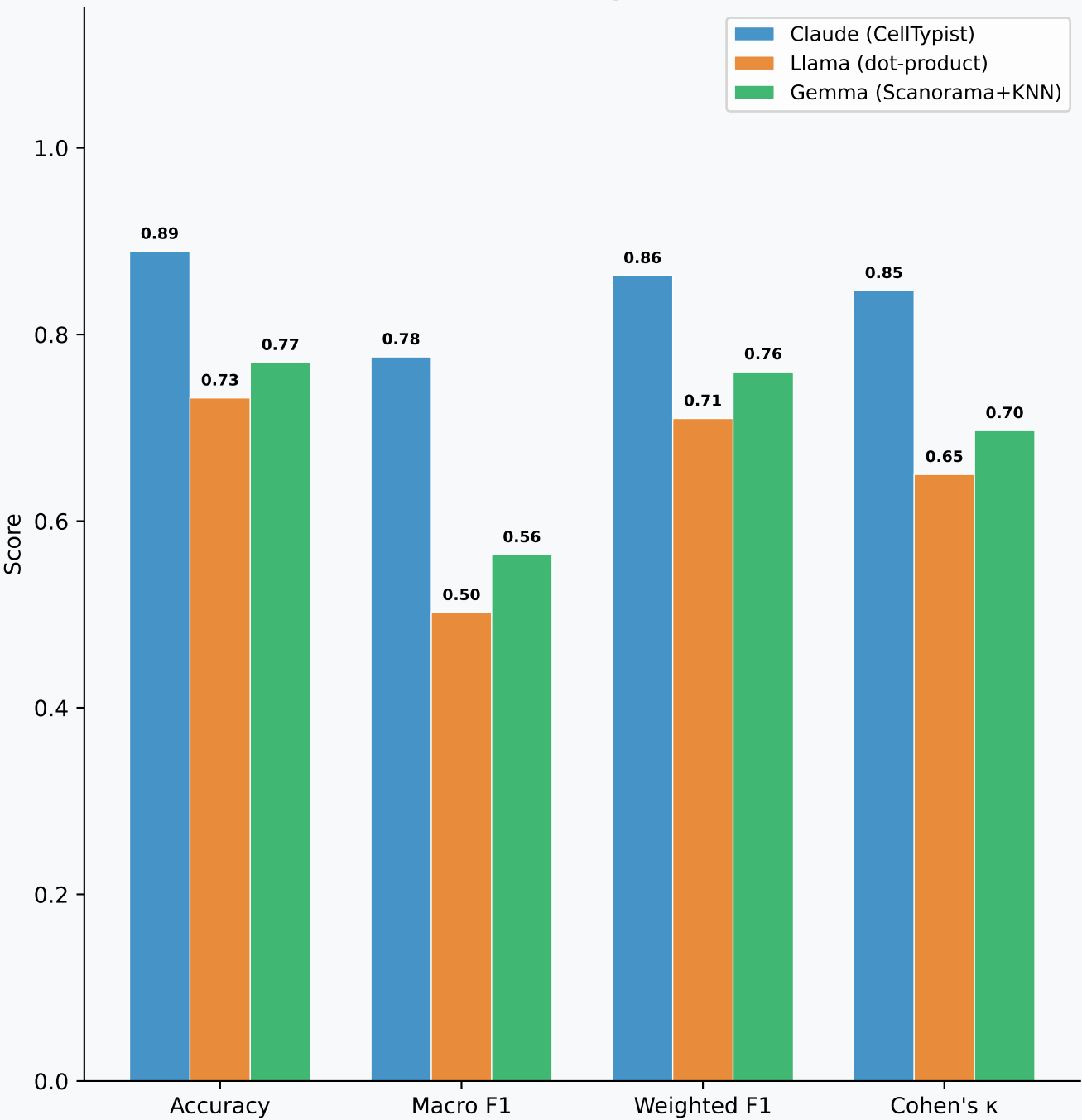
Performance Summary

Model	Accuracy	Macro F1	Wtd F1	Kappa	Time	Mem
Claude (Haiku 4.5)	0.851	0.673	0.804	0.794	2.0s	10MB
Llama (3.1 70B)	0.851	0.673	0.804	0.794	1.1s	80MB
Gemma (3 27B)	0.764	0.482	0.692	0.674	23.5s	216MB

Reference Mapping Results

Annotate cells using a well-characterised external reference atlas

Metric Comparison



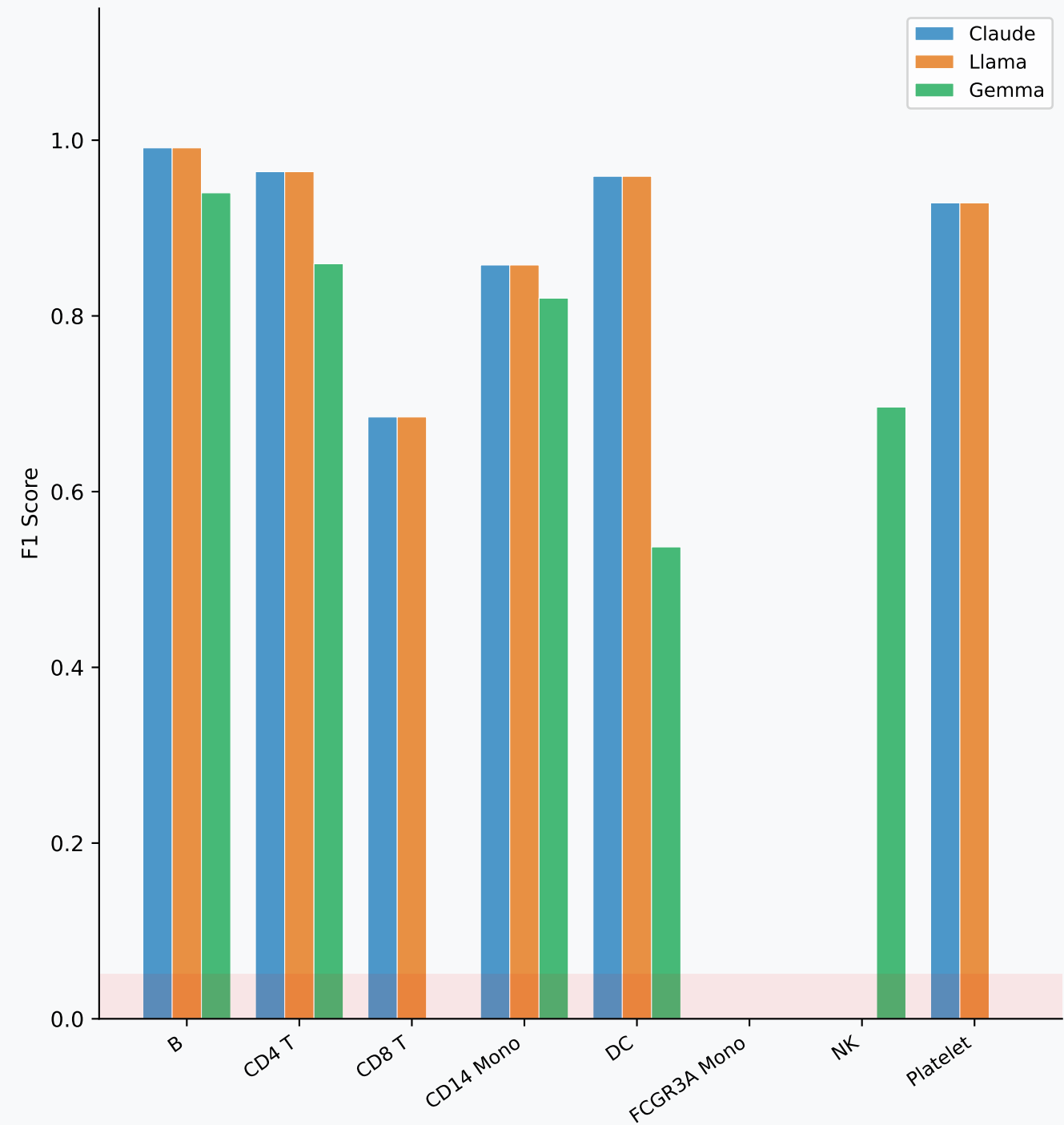
Performance Summary

Model	Accuracy	Macro F1	Wtd F1	Kappa	Time	Mem
Claude (CellTypist)	0.889	0.776	0.863	0.847	1.5s	255MB
Llama (dot-product)	0.732	0.502	0.710	0.650	0.3s	35MB
Gemma (Scanorama+KNN)	0.770	0.564	0.760	0.697	1.4s	77MB

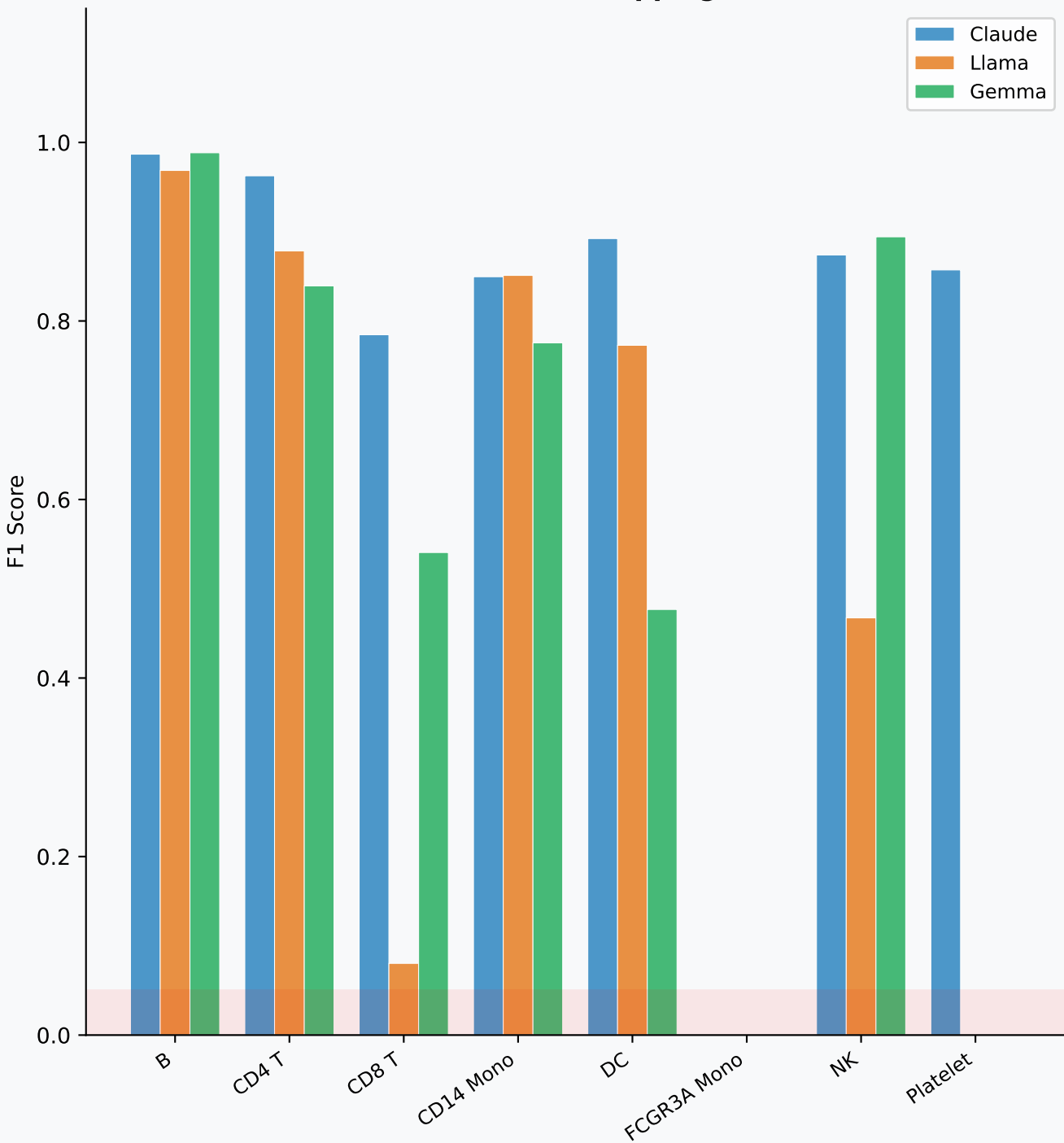
Per-Class F1 Score Breakdown

Which cell types are each method good / bad at?

Zero-Shot Annotation



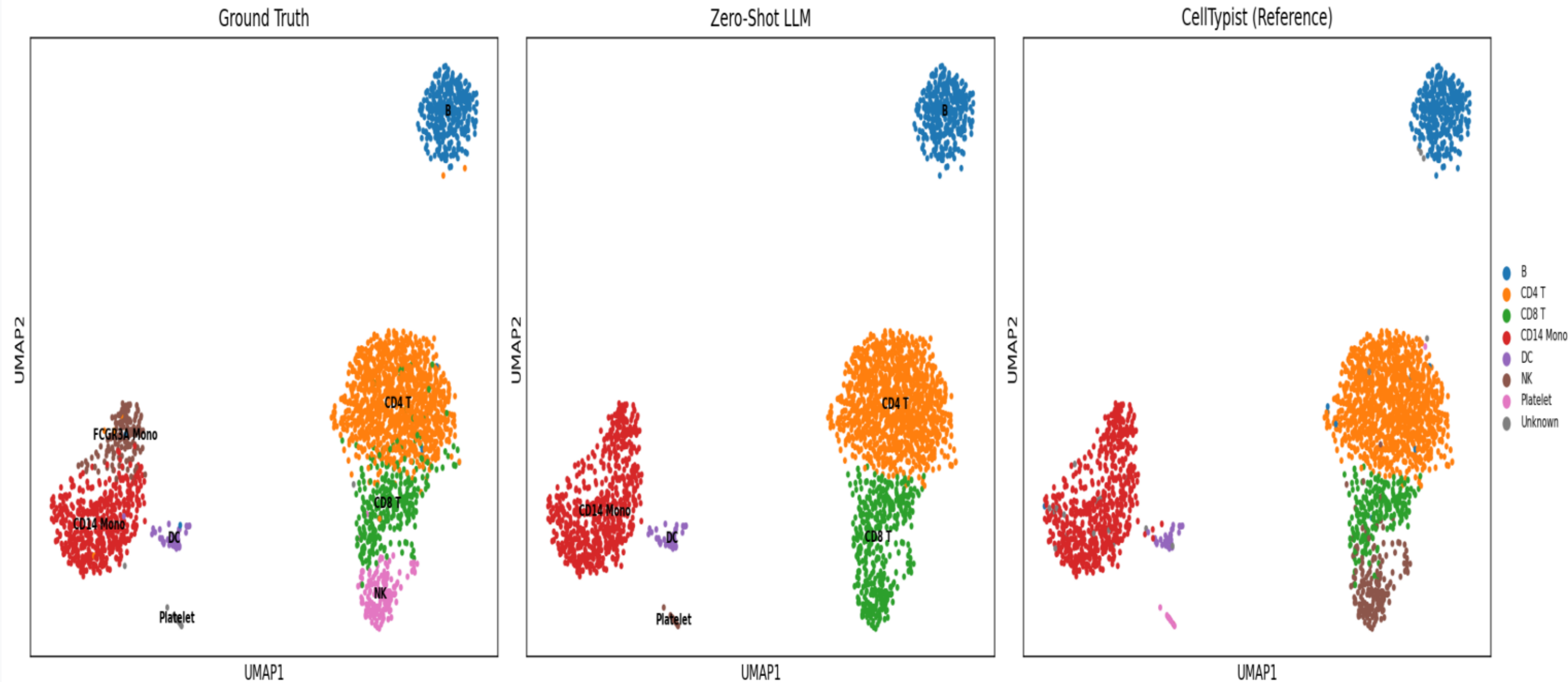
Reference Mapping



Red zone (F1≈0): FCGR3A Mono undetected by all methods — merged with CD14 Mono in the same Leiden cluster

UMAP Visualisations

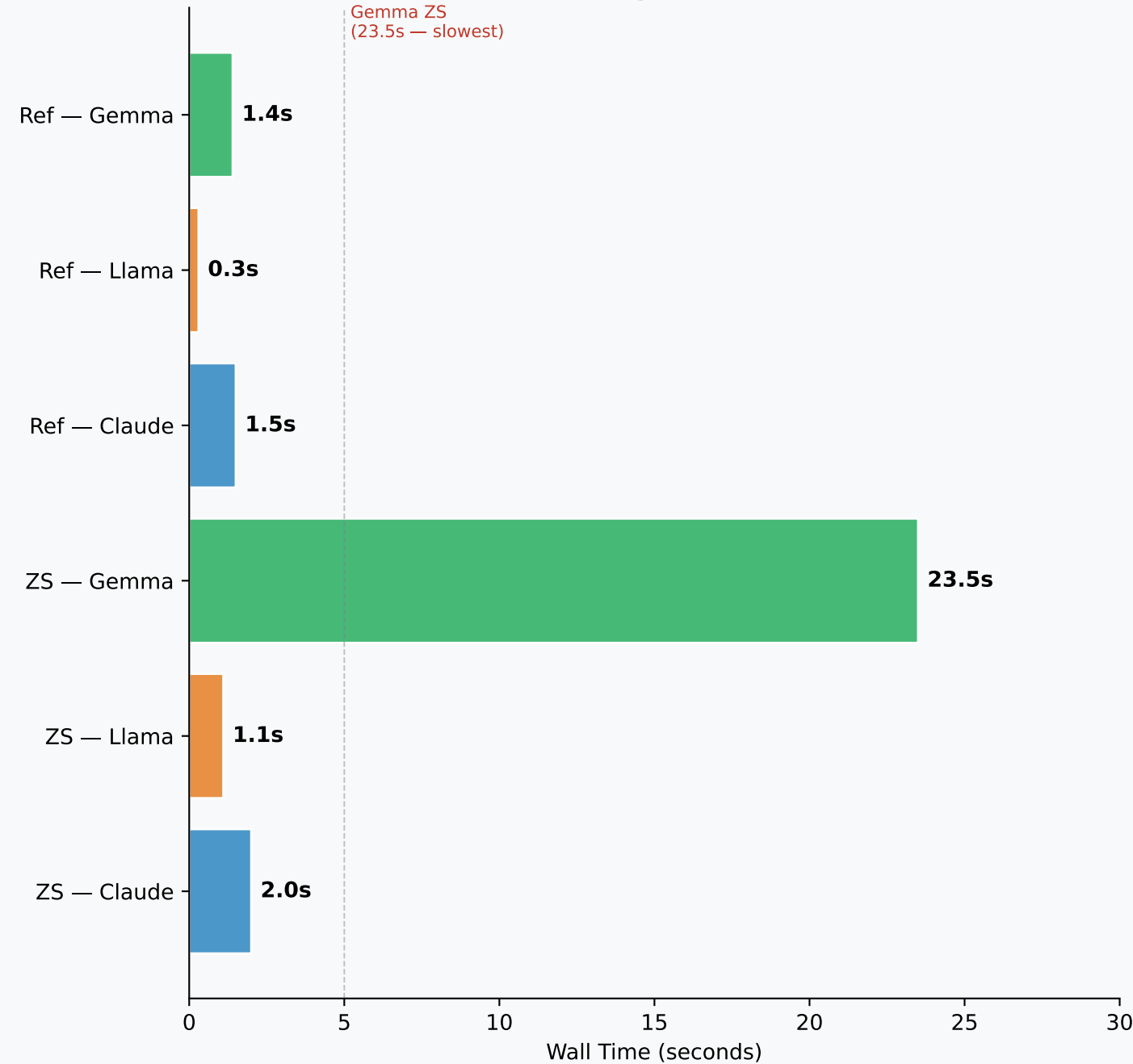
Claude pipeline — Ground Truth vs Zero-Shot vs Reference Mapping



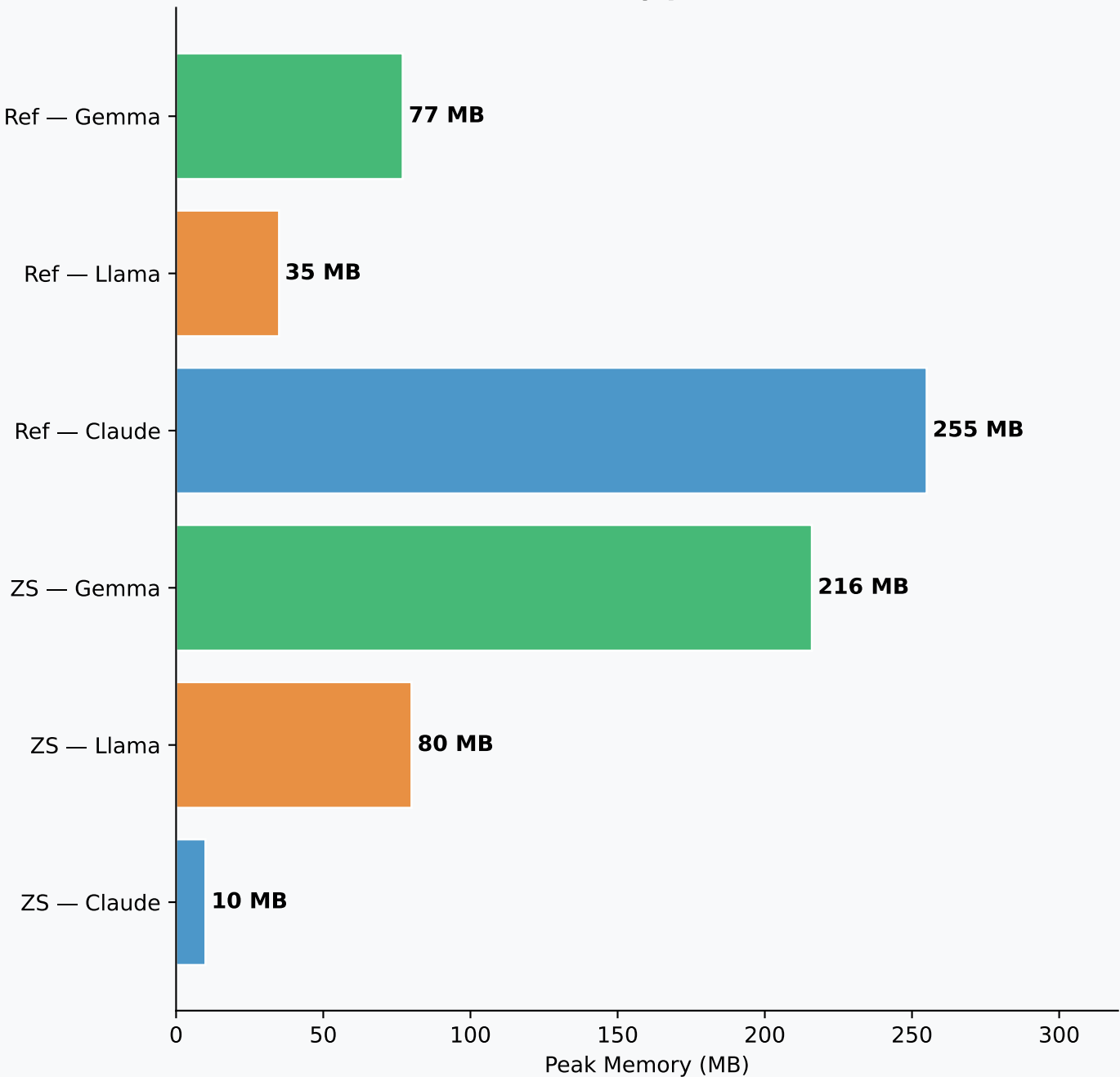
Resource Usage: Time & Memory

Wall-clock time and peak memory per annotation method

Runtime per Method



Peak Memory per Method



Lighter bars = Zero-Shot | Solid bars = Reference Mapping

Key Differences & Limitations

[1] METHOD DIFFERENCES

- Claude: Only LLM that called the Bedrock API — used biological reasoning to annotate from marker genes
- Llama: Simplified cluster-level scoring — no API call, faster, identical zero-shot accuracy to Claude
- Gemma: Suggested Scanorama (a proper batch-correction tool) — most sophisticated integration approach
- Reference tool choice drove the largest accuracy gap: CellTypist (700k cells) vs dot-product/KNN (700 cells)

[2] SHARED LIMITATIONS

- Leiden resolution 0.5 → 6 clusters, but 8 ground-truth cell types → FCGR3A Mono & NK merged with neighbors
- FCGR3A+ Monocytes: F1 = 0.00 for ALL methods — requires higher clustering resolution to rescue
- pbmc68k reference (Llama/Gemma) too small (700 cells) for reliable centroid estimation
- Ground truth labels from scanpy (coarse: 'CD4 T cells') vs Seurat (fine: 'Naive/Memory CD4 T')

[3] LLM CODE QUALITY

- Claude: Produced complete, runnable pipeline — chose CellTypist without being told about it
- Llama: Skeleton with bugs (wrong argmax logic, wrong boto3 syntax) — required ~10 fixes to run
- Gemma: Most structured (6 modular files) but wrong reference (Mouse Brain Atlas for a PBMC task!)